

Water Electrolysis

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Split water and see a 2:1 ratio

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 4 — Water Electrolysis and Gas Collection

Hook: Tap water barely splits — add electrolyte, measure current, prove 2:1 stoichiometry, then pop an H₂ balloon.

Learning outcomes

- Explain why **tap water alone** electrolyzes poorly (few ions)
- Show that **Na₂SO₄** electrolyte increases current and reaction rate
- Measure **current** and estimate **H₂/O₂ production rate** with Faraday's law
- Relate applied **voltage** to the ~1.23 V minimum for water splitting
- Collect gases in a **U-tube** and confirm **~2:1 H₂:O₂** stoichiometry
- Identify H₂ with an instructor-led **balloon pop** demo
- **Start** electrolytic rust-removal cells for Session 5

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: will graphite in tap water make gas? Why so slow?	lecture.md
10–20 min	Electrolyte, half-reactions, Faraday rate, 2:1 preview	lecture.md
20–30 min	Part A — Beaker + tap water only; record tiny current	experiment.md — Part A
30–45 min	Part B — Add Na ₂ SO ₄ ; measure I; calculate rate / discuss V	experiment.md — Part B
45–70 min	Part C — U-tube gas collection; volume ratio	experiment.md — Part C
70–80 min	Part D — Instructor H ₂ balloon fill + pop	experiment.md — Part D
80–90 min	Part E — Start derusting cells for Session 5	experiment.md — Part E

Key reactions

Overall: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (–): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (acidic media)

or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ (basic media)

Rate (Faraday): $\dot{n}(\text{H}_2) = I / (2F)$, $\dot{n}(\text{O}_2) = I / (4F)$

Status

- [] Beaker + graphite + multimeter stations ready
- [] ~0.1 M Na₂SO₄ mixed and labeled

- [] U-tube apparatus pilot-tested (2:1 visible)
- [] Balloon demo rehearsed or cancelled per venue rules
- [] **No saltwater** — Na_2SO_4 only for electrolysis
- [] Rusty nails + washing soda ready for Part E

Session 4 — Lecture: Water Electrolysis and Stoichiometry

Target duration: ~15–20 minutes (concepts interleaved with Parts A–B work)

Instructors: electrolyte contrast, Faraday rate, balloon safety — instructor.md.

Opening hook (0–10 min)

Show two graphite rods in a beaker of **tap water**.

"If water is H_2O , why doesn't electricity immediately split it into visible gas? What are we missing?"

Preview today's arc: slow → add electrolyte → measure current/rate → collect volumes → H_2 balloon pop.

Core concepts (10–20 min)

1. Why tap water is a poor electrolyzer

- Pure / tap water has **very few ions**
- Current needs a closed path of mobile charge carriers
- Result: tiny current, almost no bubbles (**Part A**)

2. Role of sodium sulfate

- Add **Na_2SO_4** → Na^+ and SO_4^{2-} carry current
- Supporting electrolyte: enables conduction; water is still what is split
- **Never NaCl** (chlorine risk)
- Result: larger current, fast bubbling (**Part B**)

3. Voltage needed

- Thermodynamic minimum $\approx$ **1.23 V** to split water
- Real cells need **more** (overpotential + resistance)
- Students record the **applied voltage** they actually use

4. Rate from current (Faraday)

$$\dot{n}(H_2) = I / (2F) \quad \dot{n}(O_2) = I / (4F)$$

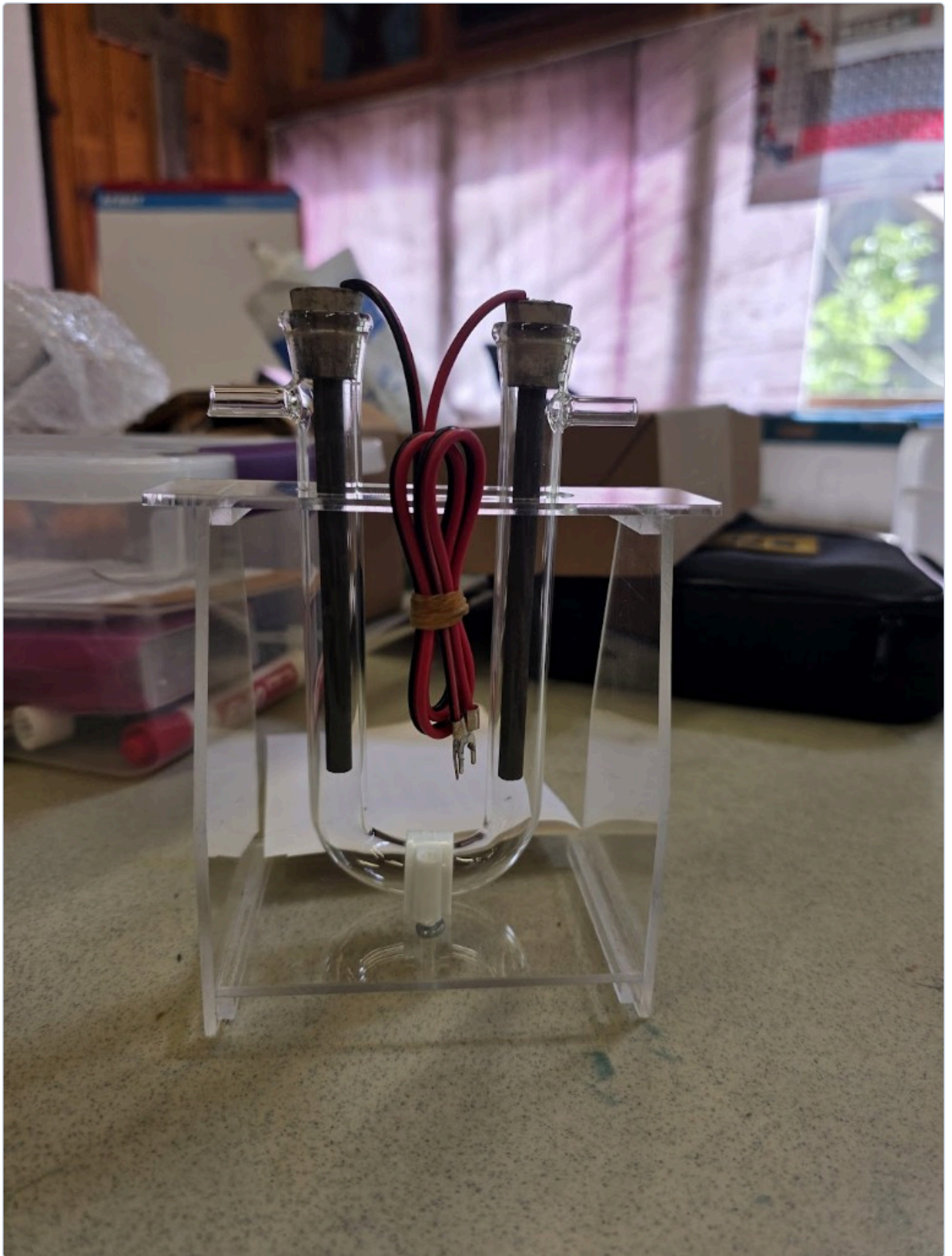
$$F \approx 96485 \text{ C/mol } e^-$$

Same current $\Rightarrow H_2$ mole rate is **twice** O_2 mole rate.

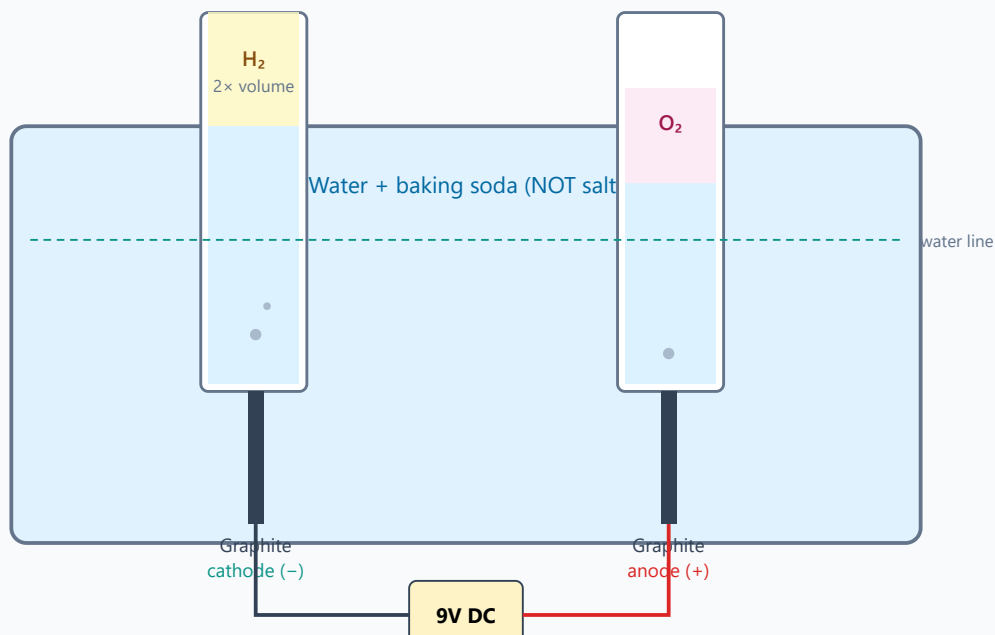
5. Stoichiometry in volumes (Part C)



U-tube collects the two gases separately so students can measure the ratio.



Session 4 - Water electrolysis + gas collection



Fill tubes completely with water, invert underwater - measure gas volume over time

Expected ratio $V(\text{H}_2) : V(\text{O}_2) \approx 2 : 1$

6. Identifying hydrogen (Part D)

Instructor collects cathode gas in a **balloon** and ignites → pop. Combustible = H_2 .

Energy discussion (if time after Part D)

1. Extra renewable electricity → electrolysis → H_2
2. Store H_2 ; later use in a fuel cell → electricity + water
3. H_2 is an **energy carrier**, not a primary source

Preview Session 5 (end of class)

"We used electricity to make hydrogen. Overnight we'll use electricity to undo rust. Tomorrow: corrosion, galvanic cells, and sacrificial protection."

Start derusting cells per experiment.md Part E.

Vocabulary

- [] Electrolyte / supporting electrolyte
- [] Electrolysis
- [] Faraday's law (rate from current)
- [] Stoichiometry / 2:1 volume ratio

- [] Energy carrier vs. energy source
-

Safety talking points

- **Never use saltwater** — chlorine gas risk
- Ammeter in **series**
- Balloon pop / flame: **instructor only**, small balloon, clear area

Session 4 — Instructor Notes: Water Electrolysis Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Students have plated metals. Today they discover that **splitting water needs ions in solution**, quantify **rate from current** (Faraday), prove **2:1 stoichiometry** by gas collection, and see a dramatic **H₂ balloon pop**. Overnight electrolytic rust removal still starts at the end for Session 5.

Teaching sequence (keep this order)

1. **Tap water beaker** — expect disappointment (almost no bubbles). That disappointment is the lesson.
2. **Add Na₂SO₄** — same voltage → much larger current → fast bubbles. Measure I.
3. **Calculate** $\dot{n} = I/(nF)$ and discuss why applied $V > 1.23$ V.
4. **U-tube volumes** — same 2:1 that Faraday already predicted.
5. **Balloon** — qualitative ID of H₂ (instructor only).
6. **Derusting cells** — handoff to Session 5.

Atomic / molecular foundations

Why tap water fails

- Autoionization: $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ with $[\text{H}^+][\text{OH}^-] = 10^{-14}$ at 25 °C — vanishingly few ions.
- Tap water adds some minerals, but still a **poor conductor** compared with 0.1 M Na₂SO₄.
- Without current, half-reactions barely run → “slow reaction.”

Why Na₂SO₄ works

- Dissolves to Na⁺ and SO₄²⁻ — **supporting electrolyte**.
- Water remains the species oxidized/reduced (classroom story).
- **Never NaCl** → Cl₂ risk at the anode.

Electrolyte	OK?	Why
Sodium sulfate	Yes — planned	Inert supporting electrolyte
Baking soda	Backup only	No chlorine
Table salt NaCl	NO	Cl ₂ gas

Half-reactions and the 2:1 ratio

Cathode (–): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (or base form with OH^-)

Overall: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

Electron balance: 4e^- make **2 H₂** and **1 O₂** $\Rightarrow$ volume ratio 2:1 at same T, P.

Faraday coaching (Part B)

$$\begin{aligned}\dot{n}(\text{H}_2) &= I / (2F) & \dot{n}(\text{O}_2) &= I / (4F) \\ F &\approx 9.65 \times 10^4 \text{ C/mol}\end{aligned}$$

- Students must convert **mA** $\rightarrow$ **A** before dividing.
- Emphasize: **same I** already implies $\dot{n}(\text{H}_2) = 2 \dot{n}(\text{O}_2)$ — volumes later confirm chemistry, not a new idea.
- Optional volume rate: use ~ 24.5 L/mol at $\sim 25^\circ\text{C}$ (or 22.4 L/mol at STP — pick one and stay consistent).

Voltage talking points

- Thermodynamic minimum $\approx$ **1.23 V**.
- Graphite + solution resistance + overpotentials $\rightarrow$ classroom cells often need **several volts**.
- “Voltage needed” = discuss 1.23 V floor + record **applied V**; if supply is adjustable, hunt the onset of visible bubbling.

Pilot tip: note typical I (tap) vs I (Na_2SO_4) at your fixed V so TAs know what “good” looks like.

U-tube coaching (Part C)

Glass U-tube, graphite through stoppers, side arms vented, Na_2SO_4 fill. Black (–) = faster H_2 limb. Accept ratio **1.6–2.2**.

Ratio error cause	Effect
O_2 more soluble	Ratio high
Escaped bubbles / leaks	Skew either way
Air at $t = 0$	Systematic error

H₂ balloon pop (Part D) — safety first

This is a **demo**, not a free-for-all.

1. Feed **cathode gas only** into a **small** balloon (anode vents separately).
2. Modest inflation — not a large balloon.
3. Tie off; move away from the bath and faces.
4. Long lighter; arm’s length; clear space; extinguisher ready.

5. Skip if venue forbids open flame.

Never inflate with mixed H_2/O_2 . **Never** use a rigid sealed vessel.

Loud pop startles — warn the room; protect ears if needed.

Part E — Rust removal handoff

Role	Connection
Rusty iron	Battery – (cathode)
Sacrificial steel/graphite	Battery + (anode)
Electrolyte	Washing soda or baking soda — not NaCl

Photograph before; control nail without power; overnight policy confirmed.

Misconceptions

Misconception	Correction
“Bubbles are steam.”	H_2 and O_2 gases, not boiling.
“Salt makes it better.”	Salt makes chlorine — forbidden.
“We need huge voltage because water is special.”	Minimum $\sim 1.23\text{ V}$; extras are kinetics/resistance.
“Current and stoichiometry are unrelated.”	Faraday is the stoichiometry in electron language.
“Hydrogen is a fuel like oil.”	It’s a carrier ; energy came from electricity.

Board plan

1. Sketch beaker + two graphite rods; label “few ions.”
2. Add Na_2SO_4 ; write $\text{I}\uparrow \rightarrow \text{rate}\uparrow$.
3. Write $\dot{n} = I/(nF)$ and 1.23 V note.
4. Write $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$; circle 2:1.
5. Safety: no NaCl; balloon = instructor only.
6. Start derusting before dismissal.

Optional enrichment

- Conductivity of tap vs $0.1\text{ M Na}_2\text{SO}_4$ (if you have a meter).
- Overpotential on graphite vs platinum (names only).
- Industrial alkaline vs PEM electrolyzers (names only).

Session 4 — Preparation

Before class (instructor)

- [] Pilot **Part A** → **B** in a beaker: record I (tap) vs I (Na_2SO_4) at fixed V
- [] Pilot Faraday arithmetic with those currents (students will copy your method)
- [] Pilot **U-tube** — confirm 2:1 ratio visible within session time
- [] Rehearse **H₂ balloon** demo (cathode-only feed, small balloon) or cancel per venue rules
- [] Check graphite rods, stoppers, spade connectors, multimeters
- [] Mix **~0.1 M Na₂SO₄** stock; label **"NO SALT"**
- [] Pre-weigh Na_2SO_4 portions (~3.5 g) for beaker additions if helpful
- [] Stage rusty nails, washing soda, sacrificial anodes, jars for **Part E**
- [] Confirm overnight power policy for derusting cells

Day-of setup

- [] Beaker stations: graphite, leads, series ammeter ready
- [] U-tube stations filled or ready to fill with Na_2SO_4
- [] Large sign: **"No salt in electrolysis — use sodium sulfate"**
- [] Balloon demo area marked; extinguisher present; long lighter staged
- [] Part E station ready for last 10 minutes

Common failure modes (from pilot)

Issue	Prevention
Part A "broken" complaint	Brief beforehand: tiny current is the point
Ammeter reads zero after Na_2SO_4	Wrong range; meter not in series; bad clip
Students grab salt	Pre-brief; hide NaCl; labeled Na_2SO_4 only
Balloon mixes gases	Dedicated cathode tube; anode vents free
Overpressure in U-tube	Side arms open

Open questions / notes

- Fixed voltage used for A/B comparison:
- Typical I_{tap} / $I_{\text{Na}_2\text{SO}_4}$:
- Balloon demo yes/no for this venue: